

Lab 1

The screenshot shows a PostgreSQL query editor with the following SQL query:

```
1 SELECT relname as object_name,
2     pg_size_pretty(pg_table_size(fed.oid)) as data_size,
3     pg_size_pretty(pg_indexes_size(fed.oid)) as index_size,
4     pg_size_pretty(pg_total_relation_size(fed.oid)) as total_size
5 FROM pg_class fed
6 JOIN pg_namespace n ON n.oid = fed.relnamespace
7 WHERE fed.relkind = 'r' AND n.nspname = 'public'
8 ORDER BY pg_total_relation_size(fed.oid) DESC;
```

The Data Output tab shows the results of the query:

	object_name name	data_size text	index_size text	total_size text
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	216 kB	952 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	city	64 kB	48 kB	112 kB
9	film_category	72 kB	40 kB	112 kB
10	actor	40 kB	32 kB	72 kB
11	store	8192 bytes	32 kB	40 kB
12	staff	16 kB	16 kB	32 kB
13	language	8192 bytes	16 kB	24 kB

Total rows: 15 | Query complete 00:00:00.284 | CRLF | Ln 8, Col 47

Lab 2

/* current size */

The screenshot shows a PostgreSQL query editor with the following SQL query:

```
10 /* current size */
11 SELECT pg_size_pretty(pg_indexes_size('payment'));
12 /* create heavy index on payment_date */
13 CREATE INDEX idx_payment_date_long
14 ON payment(payment_date, amount, customer_id);
15 /* check size after create index */
16 SELECT pg_size_pretty(pg_indexes_size('payment'))
```

The Data Output tab shows the results of the query:

	pg_size_pretty text
1	920 kB

Successfully run. Total query runtime: 70 msec. 1 rows affected.

Total rows: 1 | Query complete 00:00:00.070 | CRLF | Ln 11, Col 51

/* create heavy index on payment_date */

```
10 /* current size */
11 SELECT pg_size_pretty(pg_indexes_size('payment'));
12 /* create heavy index on payment_date */
13 CREATE INDEX idx_payment_date_long
14 ON payment(payment_date, amount, customer_id);
15 /* check size after create index */
16 SELECT pg_size_pretty(pg_indexes_size('payment'))
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 60 msec.

✓ Query returned successfully in 60 msec. ✕

Total rows: Query complete 00:00:00.060 CRLF Ln 14, Col 47

/* check size after create index */

```
10 /* current size */
11 SELECT pg_size_pretty(pg_indexes_size('payment'));
12 /* create heavy index on payment_date */
13 CREATE INDEX idx_payment_date_long
14 ON payment(payment_date, amount, customer_id);
15 /* check size after create index */
16 SELECT pg_size_pretty(pg_indexes_size('payment'))
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	pg_size_pretty text
1	1488 kB

✓ Successfully run. Total query runtime: 42 msec. 1 rows affected. ✕

Total rows: 1 Query complete 00:00:00.042 CRLF Ln 16, Col 51

Discussion: New index consumes $(1488 - 920)/8 = 71$ pages

Lab 3

/* Create partial index */

```
20 /* Create partial index */
21 CREATE INDEX idx_expensive_films
22 ON film(replacement_cost)
23 WHERE replacement_cost > 25.00;
24
25 /* Find size of this specific index */
26 SELECT relname, pg_size_pretty(pg_relation_size(relid))
27 FROM pg_stat_user_indexes
28 WHERE relname = 'idx_expensive_films';
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 57 msec.

Total rows: Query complete 00:00:00.057 CRLF Ln 23, Col 32

/* Find size of this specific index */

```
20 /* Create partial index */
21 CREATE INDEX idx_expensive_films
22 ON film(replacement_cost)
23 WHERE replacement_cost > 25.00;
24
25 /* Find size of this specific index */
26 SELECT relname, pg_size_pretty(pg_relation_size(oid))
27 FROM pg_class
28 WHERE relname = 'idx_expensive_films';
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	relname name	pg_size_pretty text
1	idx_expensive_fil...	16 kB

✓ Successfully run. Total query runtime: 62 msec. 1 rows affected. ✕

Total rows: 1 Query complete 00:00:00.062 CRLF Ln 28, Col 39

/* Create full index on replacement_cost */

```
30  /* Create full index on replacement_cost */
31  CREATE INDEX idx_full_replacement_cost
32  ON film (replacement_cost);
33
34  /* Find size of this full index */
35  SELECT relname, pg_size_pretty(pg_relation_size(oid))
36  FROM pg_class
37  WHERE relname = 'idx_full_replacement_cost';
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 70 msec.

✓ Query returned successfully in 70 msec. ✕

Total rows: Query complete 00:00:00.070

CRLF Ln 32, Col 28

/* Find size of this full index */

```
19
20  /* Create partial index */
21  CREATE INDEX idx_expensive_films
22  ON film(replacement_cost)
23  WHERE replacement_cost > 25.00;
24
25  /* Find size of this specific index */
26  SELECT relname, pg_size_pretty(pg_relation_size(oid))
27  FROM pg_class
28  WHERE relname = 'idx_expensive_films';
29
30  /* Create full index on replacement_cost */
31  CREATE INDEX idx_full_replacement_cost
32  ON film (replacement_cost);
33
34  /* Find size of this full index */
35  SELECT relname, pg_size_pretty(pg_relation_size(oid))
36  FROM pg_class
37  WHERE relname = 'idx_full_replacement_cost';
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	relname name	pg_size_pretty text
1	idx_full_replacement_c...	40 kB

✓ Successfully run. Total query runtime: 54 msec. 1 rows affected. ✕

Total rows: 1 Query complete 00:00:00.054

CRLF Ln 37, Col 45

Task: 16kB of this specific index is small than 40kB of hypothetical full index on replacement_cost.